

Dynamic Assessment of Bilingual Children: A Guided Tutorial

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Purpose of the project

Our goal in the creation of this tutorial is to provide a comprehensive and accessible guide to dynamic assessment, particularly for culturally and linguistically diverse (CLD) children. Throughout our graduate program, we found that dynamic assessment was frequently discussed and read about in classes, but we found the transfer to clinical practice was challenging. It is our hope that this tutorial makes the practice of dynamic assessment more accessible to graduate students and practicing clinicians, through the use of a literature review, step-by-step guide, and five video case examples. Although we are particularly interested in the application of dynamic assessment to CLD populations, we believe that it can be used with any population, and administered by monolingual speech-language pathologists.

What is Dynamic Assessment?

Simply put, dynamic assessment is an interactive approach to conducting assessments within the domains of psychology, speech and language or education, which focuses on the ability of the learner to respond to intervention. Although its application has varied widely among researchers and theorists, the dynamic assessment model adopts three central tenets: its interactive nature, its focus on metacognitive learning processes and its ability to inform the clinician about the responsiveness of a learner to an intervention approach (Lidz, 144). While it can take a variety of forms, dynamic assessment is commonly administered according to a test-teach-retest paradigm, in which the child is being assessed according to their independent performance, response and engagement with intervention and overall change.

Theoretical Background

The theoretical basis for dynamic assessment was born out of the psychology literature, particularly L.S. Vygotsky and Reuven Feuerstein. Vygotsky has been referred to as the “father of dynamic assessment” (Lidz, 1995). He was critical of the approach of Western psychometric tests, arguing that one cannot make assumptions on the ability of a child based on independent and decontextualized performance. Vygotsky believed that there were two diverging types of assessment: symptomatic and diagnostic. Symptomatic assessment focused on what a particular child could do as compared to developmental or age-matched norms, whereas diagnostic assessment relied on determining what dynamic and genetic connections were defining that individual’s mental development.

Vygotsky argued that unaided performance overlooked subtle differences in cognitive functioning that can be identified by assessing how they respond to help. This help, from either adults or more capable peers, allows professionals to see which skills are established, and which skills may be emerging. This emphasis on emerging function led Vygotsky to adopt the widely used ‘zone of proximal development’ (ZPD). Essentially, ZPD asserts that learners can be characterized both by what they perform independently as well as how well they perform with the help of a mediator. The skills within a child’s ZPD are emerging, and may require adult mediation to elicit. Dynamic assessment attempts to elicit the ZPD, while static assessments elicit the zone of actual development (ZAD).

Another major contribution from Vygotsky is The Social Cultural Theory of Mind. This theory supported the notion that learning was an interactive process between the mediator and the learner. He argued that cognitive skills development was influenced and shaped by our social environment and that instruction needed to be collaborative to facilitate growth. This led to the inclusion of transcendence as a central tenet of dynamic assessment.

Feuerstein’s contribution to the theory and practice of dynamic assessment surrounded his emphasis on child motivation and transcendence as part of a mediated learning experience (MLE). Transcendence involves cooperation between the examiner/mediator and the child, in which the new skill being learned is applied to that child’s personal life and the two participants work together to apply that skill to increasingly complex and relevant tasks. Feuerstein also highlighted the importance of child motivation during mediated learning experiences, even cautioning against curriculum-based assessments

because of the negative emotions surrounding school for many children. Feuerstein also popularized the term dynamic assessment in the 1960s, in conjunction with his theories about the MLE, in which the child was encouraged to plan and reflect on his or her own learning, as well as work with the mediator to determine strategies for success.

More recent viewpoints on dynamic assessment by Haywood and Lidz argue for a transactional perspective on mental functioning, in which intelligence, cognition and motivation overlapping to determine ability. While they argue that intelligence reflects innate ability, cognitive skills and processes can be taught, and intrinsically motivated individuals are the most responsive to teaching. Therefore, dynamic assessment is based on the fact that cognitive processes are modifiable and mediated learning experiences are structured to support intrinsic motivation during the assessment through the use of transcendence, collaborative problem solving and the resistance to rely on an extrinsic reward.

Components of Dynamic Assessment:

Dynamic assessment can take a variety of forms. First, it can involve a restructuring of the testing situation, which is what you will see in our demonstration of the dynamic assessment of phonology. Dynamic assessment can also involve “learning within the test.” The easiest way to think about this is through the administration of a static assessment. One can administer a test item, explain it to the child, and then retest in the same session. The third type of dynamic assessment involves the test-teach retest paradigm and metacognitive mediation sessions. In this model, the child is given a pretest, and skills they did not demonstrate during testing are targeted in mediated learning experiences. Following these interventions, the child is given a posttest to assess their transfer of the skill. In this model, the child or client is assessed both on their change from pre to post test, but also on their participation in the mediated learning experiences.

How does Dynamic Assessment Differ from Other Models of Assessment and Intervention?

Standardized assessment

Standardized tests are conducted in such a way that the examiner or clinician provides no support to the child. They aim to measure the child’s zone of actual development, thus telling us what a child has already learned. Standardized tests do not tell us anything

about a child's learning potential. However, we do support the idea of using standardized tests as part of your assessment package. For example, we believe that standardized assessments are a good way to get a global view of a child's independent performance for a particular component of communication. They can also be a valuable tool in selecting targets for dynamic assessment.

Any "active" assessment?

This is a more subtle distinction to make. While dynamic assessment is active, it also operates under a specific theoretical rationale. An active assessment that does not focus on the facilitation of metacognitive awareness is not dynamic assessment. Similarly, without the incorporation of transcendence and the support of intrinsic motivation, the assessment cannot be considered a true dynamic assessment. Additionally, although dynamic assessment does rely more heavily on clinical skill and individualized intervention, there are guidelines for measuring learning performance and child modifiability to ensure both validity and consistency.

Response to Intervention

Response to Intervention (RTI) is an intervention system employed within an education setting in which students are continually assessed and given varying levels of support based on their ongoing performance. It has most often been studied in the realm of literacy development in young children. Although RTI does have a lot of advantages, it is not equivalent to dynamic assessment because it is still a search for deficits as opposed to potential areas of strength. Also, RTI is a specific set of procedures designed for classification according to Tiers. Dynamic assessment is not used to place people into categories, rather it is frequently used to remove them from categories (ie: intellectual disability).

Stimulability Testing

Stimulability testing, or the use of cues and prompts to ascertain whether a child is capable of producing a certain sound, is similar to dynamic assessment and widely used by clinicians. However, it is not common practice to assess whether a child is stimutable for a sound according to a systematic protocol. Do you simply model the sound, or add placement, tactile or prolongation cues? Do you account for the linguistic complexity in which the sound is being produced? How is it measured? All of the above questions can

be answered by using a standardized and systematic protocol such as the Glaspey Dynamic Assessment of Phonology (GDAP) which we discuss later.

Why do we use Dynamic Assessment?

There are many advantages to using dynamic assessment during an evaluation. Combining this thorough assessment approach with traditional testing measures, including observations, interviews and a developmental history, gives the clinician a more complete profile of the individual they are working with. The steps taken during a dynamic assessment inform intervention planning to make the most efficient use of therapy. The interactive, naturalistic approach of dynamic assessment and emphasis on motivation and transcendence makes testing less of a burden on the child. Additionally, changes over time can be observed through the use of the test-teach-retest paradigm and targeted skills can be evaluated in more detail than traditional standardized tests allow.

Shortcomings of standardized tests

Standardized assessments are completed in one session which does not provide the opportunity to observe changes over time. Many aspects important to learning and predicting a child's responsiveness to therapy are not assessed by standardized tests. Dynamic assessment reveals a child's zone of proximal development and allows the clinician to systematically reflect on the child's attention, metacognition, tolerance to failure, modifiability, and gain an understanding of what supports and motivations are most effective. If only standardized assessment tools are used a clinician can spend vital time in therapy working on goals that are above or below the child's zone of proximal development.

The application of standardized tests to CLD populations

Administering most norm-referenced standardized tests on CLD children and reporting the scores is unethical considering the vast majority of standardized tests have been normed on monolingual, non-CLD populations. Static assessments normed on English-speaking children cannot be applied to this population and retain their sensitivity and specificity.

The culture of the normed population is often reflected in the content of standardized tests. A child's score can be skewed if they ask questions that are not culturally neutral on

a standardized test. One example is in the test for Assessing Semantic Skills through Everyday Themes (ASSET). This assessment contains jargon, and lacks cultural neutrality. The test-taker is instructed to look at a picture of a gas station and point to something that has a “nozzle”, identify a “tire gauge” and define what a tire is made of. Later in the test the child is asked to describe what a “stethoscope” is used for, and to locate where the “x-ray” and “reflex hammer” are in a picture of a Western doctor's office.

While much of the language in this test is dated, considering it was published in 1988, the test content assumes that the examinee has been to a gas station, implying their family has a car, and that they have been to a doctor frequently enough to know specialized terms for medical instruments. This was not the reality of many CLD families in the late 80s, nor is it the case today. Poor performance by a CLD child cannot be fully attributed to a language learning difficulty on a test such as the ASSET, as their performance will be influenced by their class and lack of exposure to first world, Western culture.

The Goldman-Fristoe, one of the most commonly used articulatory assessments, has a more subtle Western cultural bias. One example is that the test-taker is asked to name a “plane” when presented with a visual prompt. Children from low SES families often have not been on a plane before. While this is not as relevant on an articulation test as it would be on a vocabulary test, the child's familiarity with the testing items can still influence performance.

There has been a movement within speech language pathology to improve assessment measures for CLD populations and many large testing companies have responded by translating their most widely used standardized measures into other languages. Clinicians should note that merely because a test has been translated into another language does not necessarily mean the same concepts are being tested; direct translation of a test question does not guarantee that it is eliciting the same knowledge as the original question, and it likely is not considering the complexity of language.

An example of this is that in Spanish there is more than one form of past tense: preterite and imperfect. A direct translation of the CELF into Spanish does not comprehensively evaluate a child's knowledge of the nuanced differences between these tenses. Another example is that the subjunctive is a tense that does not exist in English. A direct

translation of the CELF-English would not systematically evaluate a child's understanding of subjunctive.

Perhaps the principal concern regarding the application of translated assessment tools to CLD populations is that indicators of disorder are not universal across languages. For example, a child's understanding of present progressive and past tense is often examined in English. In Spanish, however, errors in clitic pronouns are a sign of disorder (Restrepo & Gutiérrez-Clellen, 2001), not the use of -ando, -iendo, or past tense. A direct translation from English to Spanish would not capture this crucial difference.

We do not completely dismiss the use of standardized assessment; they should be a component of the assessment battery. Norm-referenced standardized measures are excellent tools for collecting pre and posttest data of a particular skill during a dynamic assessment. The clinician must always be cautious about comparing a child's posttest performance to test norms. The nature of this assessment de-validates comparison to the normative sample.

Differentiating difference from disorder

Differentiating a communication disorder from a difference is paramount to ensuring accurate evaluation and classification of CLD populations. The evolving demographics of this country demand that SLPs be competent in making this crucial distinction. Culturally and linguistically diverse children are increasingly being referred for speech and language evaluations and dynamic assessment should be utilized when making diagnoses and clinical decisions.

Empirical support

The most powerful evidence to support dynamic assessment comes from the literature's support of this approach. Throughout this handout, we reference particular studies that confirm the validity of dynamic assessment as an evaluation tool. Below we have listed some of the most well-known studies, specifically those that apply to bilingual populations:

- Olswang and Bain (1996) demonstrated in their study that dynamic assessment was more accurate than standardized measures at predicting immediate language growth.

- Modifiability has been shown to be effective in differentiating children with low language ability from their typically developing peers (Peña, 2000).
- Kapantzoglou, Restrepo, and Thompson demonstrated in 2012 that dynamic assessment was more effective at differentiating children with typical language development from children with primary language impairment.
- Gillam and Peña in 2004 found that classification accuracy of CLD children reached 100% when clinician judgement was combined with pre-test to posttest change.
- Gray et.al (vocabulary testing) indicated that dynamic assessment of word learning better identifies lexical skills of children with language impairments, as compared to standardized tests.
- Peña (2000) found that modifiability (planning, attention to task, motivation, transfer, responsivity, and examiner effort) was more effective in differentiating children with low language ability from typically developing children with CLD backgrounds.

Dynamic Assessment in Practice:*Children from Culturally and Linguistically Diverse Backgrounds*

As described in the above section, the content of standardized assessments is rarely, if ever, culturally neutral, and it is virtually impossible for speech-language pathologists to use the results of a standardized test to determine whether that performance was affected by cultural or linguistic differences, lack of experiences, or a genuine language impairment. Gillam and Peña (2004) determined in a literature review that simply using the pretest to posttest change led to classification accuracy of 87.1% accuracy. When the clinicians' judgment of modifiability was also considered, the accuracy increased to 100%. This led the authors to argue that "...the entire dynamic assessment process leads to outstanding classification accuracy in children from diverse ethnic backgrounds" (4).

In another study, Gray and her colleagues indicated that dynamic assessment of word learning was more accurate at identifying children with disorders than traditional vocabulary tests. Their study showed that CLD children with a language difference showed a significant increase in their posttest scores after the examiner provided appropriate mediated learning experiences. These populations also showed higher modifiability and decreased examiner effort when compared to the children who were identified as disordered (Gutierrez-Clellen & Peña, 2001).

Research has shown that effective assessment and intervention is most efficacious when administered in the child's strongest language. For many bilingual clinicians, that means approaching dynamic assessment with the added linguistic component. Taking a careful history and interviewing parents and families can help determine which language the child is most dominant in. However, simply being able to speak Spanish, Vietnamese or Russian does not automatically make you adept at conducting dynamic assessment in these languages. One must have knowledge of the developmental speech and language milestones for these languages and be able to integrate that into pretest administration and target selection. Resources exist that provide insight into the grammatical and phonological development of speakers of a variety of languages. The clinician should keep in mind that information on monolingual foreign language speakers often does not account for variations in English language learners. However, one of the most powerful components of dynamic assessment is its applicability in a variety of cultural and linguistic backgrounds, regardless of whether or not the clinician can speak the child's native language.

Resources for the SLP:

- <http://www.multicsd.org/> - This is a website created by Dr. Christina Gildersleeve-Neumann and contributed to by speech and hearing graduate students from Portland State University. It contains a wealth of information about a variety of different aspects of language and culture, including implications for the speech language pathologists. This is a great place to begin.
- For detailed information on Spanish: Bilingual Language Development and Disorders, Brian Goldstein, 2010.

Dynamic Assessment with Young Children

For children aged 2-5 it is generally not appropriate to use any sort of curriculum-based dynamic assessment paradigm, as we will encounter a high variability of child experience with “academic” instruction. In their book, Lidz and Haywood (2006) advocate for a specific assessment, The Application of Cognitive Functions Scale (Lidz, 1991) that assesses six cognitive functions representative of learning processes that are necessary for preschool programs. The six functions, or subscales, are short-term auditory memory, short-term visual memory, sequential pattern completion, perspective taking and verbal planning. All six processes are administered in a test-teach-retest format, and scoring reflects the degree of success for each task, as well as a behavior rating scale, which records self-regulation, persistence, frustration, tolerance, motivation, interactivity and responsivity.

When assessing discrete cognitive processes such as these, it is important to note that we are actually trying to modify the meta-cognitive processes around them. For example, we do not directly affect the process of attention in an impairment-based manner, but rather mediate the way in which the individual regulates that attention.

While we found the model described in Lidz and Haywood’s book a valuable tool for conducting dynamic assessment with young children, we believe that it is not the only way to approach dynamic assessment with young children. A clinician could determine their own way to assess learning readiness skills, and approach mediated learning experiences without the “semi script” provided in this assessment. However, the ACFS is a useful tool, particularly for anyone less experienced in dynamic assessment.

Dynamic Assessment with School-Aged Children

When assessing a child for the presence of a speech or language disorder, we are trying to determine whether their disorder hinders their ability to participate in their education. Dynamic assessment should then focus on something that the child is required to do in the classroom. Additionally, when the child is at school you have the ability to learn about their performance in a wide range of contexts. Starting your evaluation with a parent and teacher interview and classroom observation can help narrow your potential dynamic assessment targets.

The first step in a dynamic assessment is to select a task that will be assessed. It is important to consider the cognitive processing demands that this task may make on the learner to identify potential supports. For example, if you are trying to teach a task that requires a high level of working memory, such as the present subjunctive in Case Study 2, you want to consider whether visual supports or reminders are necessary to facilitate success.

Guidelines for Conducting Dynamic Assessment:*Target selection*

Before selecting a target, the clinician should interact with the child and talk to educators to gain an understanding of areas of difficulty for the child. Once this understanding has been achieved, a domain is selected to test. Examples of domains include receptive and expressive language, speech and cognitive processing skills.

Next, a learning target is selected within the specified domain that is deficient or emerging. Examples include assessing adjective use in the domain of language, final consonants for speech, and narrative development when analyzing cognition. The target should be developmentally appropriate and within the child's zone of proximal development. An appropriate task is something that the individual has not mastered, but is not too difficult for them to attain based on other developmental information. Targets that solely test memory capacity, such as vocabulary expansion, are not appropriate for dynamic assessment. Many children with language disorders and learning difficulties present with working-memory challenges (Boudreau & Constanza-Smith). The amount of time the child has received academic instruction in English should be taken into account as well.

Developing a protocol

While most dynamic assessments will require the clinician to create a protocol that is individualized to be sensitive to the age, ethnicity/culture and skill level of the child, some dynamic assessment protocols do exist for commonly targeted skills.

One published protocol for narrative dynamic assessment is *Dynamic Assessment and Intervention: Improving Children's Narrative Abilities*. This kit applies a test-teach-retest paradigm to narrative abilities. Two different, culturally-neutral wordless picture books are utilized for pre- and posttest narrative elicitations: *Two Friends* (L. Miller, 2000a) and *Bird and His Ring* (L. Miller, 2000b). The child is evaluated after the pretest on their inclusion or use of story components, episode structure, ideas and language. Two targets are then selected for focused instruction during two MLEs. A child's modifiability and use of targeted concepts in the posttest narrative is used to determine their learning potential. Children who have low modifiability and difficulty generalizing new narrative concepts to their stories are determined to have language-learning difficulties.

The Glaspey Dynamic Assessment of Phonology (GDAP, in press) is a scaffolding scale for dynamically assessing progress towards phonological learning targets and tracking progress over time. A cueing hierarchy is used to systematically evaluate the level of support needed to produce a given speech sound. The scale examines the sound in isolation through to connected speech, and has gradations of support from no instruction to verbal model, prolongation, segmentation and tactile cue. A child receives a score between 1 and 15 with 1 indicating that the child produced the sound without clinician support in the most advanced context, connected speech. A score of 15 indicates the child is not stimuable even with maximum support in isolation. The lower the child's score the more modifiable they are. See chart below:

Glaspey Dynamic Assessment of Phonology (GDAP)				
Cues →	Level 0 No instruction or verbal model	Level 1 Instruction & verbal model	Level 2 Instruction, Verbal model, Prolongation, Segmentation	Level 3 Instruction, Verbal model, Prolongation, Segmentation, Tactile
Environments ↓				
A. Connected Speech	1			
B. 2-target sentence	2	3		
C. 4-word Sentence	4	5		
D. 3-word Sentence	6	7	8	9
E. Word	10	11	12	13
F. Isolation				14
Not Stimuable				15

Pretest

After selecting a domain and a target the pretest is conducted. Standardized tools are an effective way to evaluate a child's existing knowledge and work well for this portion of the process. For example, when evaluating a child's knowledge of colors, utilizing the Preschool Language Scale (PLS) is an excellent way to carry out the pretest. Language sampling can also be a valuable tool for identifying potential learning targets. The clinician should research whether a standardized test exists in that child's native language before settling on an English assessment or creating one. Before beginning the clinician

should also review phonological and syntactical differences between the child's native language and English, and educate themselves on any dialectal differences that may exist (Saad, 2002).

Mediated Learning Experiences

The teaching portion of dynamic assessment is referred to as a mediated learning experience (MLE) and is conducted between the pre- and posttests. MLEs are individualized one-on-one tutorials that teach the child the targeted skill in a supported, systematic, meaningful and explicit manner. The test situation transitions from an evaluative interaction to a teaching interaction at this stage in the process (Gutierrez-Clellen & Peña, 2001). A child's ability to learn, or 'modifiability', is quantified by examining their pretest to posttest change and their receptiveness during the MLEs. Dynamic assessment posits that children who are highly modifiable and show test gains in targeted teaching areas have typical language learning abilities, while children who have low modifiability and pretest to posttest change will have difficulty learning language (Peña, Iglesias & Lidz 2001).

After analyzing the results of the pretest, the structure of the MLE is defined. Observations from the pretest and any previous interactions inform the design of the mediated learning sessions. It is essential that the mediated learning sessions reach the child on a personal level to make them understand why the skill is important for them to learn. This component of dynamic assessment is referred to as transcendence. As stated earlier, transcendence involves cooperation between the examiner/mediator and the child, in which the new skill being learned is applied to that child's personal life.

During the MLE the clinician should be explicit in conveying the intent of the MLE by clearly stating the target using concrete language. Describing what is going to be taught and how the child needs to participate helps the child create expectations. It is also important to provide the child with feedback throughout the MLE about their performance and positive reinforcement of successful application of the learning target. Elicit active conversation from the learner about their input related to the work and capitalize on all opportunities for reinforcement of transcendence.

Rating their progress

Multiple protocols for rating progress during a dynamic assessment exist; however, we elected to use the most comprehensive protocol we found in our search by Megan Stephens, MS, CCC-SLP (citation?), a practicing bilingual SLP in Portland, OR. In this protocol student responsiveness is evaluated by rating the child using a five point scale on *attention, task specific performance, planning, self-regulation/awareness, motivation and interaction with adults*. Teaching effort is quantified using a Modifiability Scale that assesses *examiner effort, student responsiveness and transfer*. Modifiability scores should be given particular attention considering they have been shown to have both high diagnostic specificity and sensitivity in identifying children with language disorder (Peña, 2000).

A total learning strategy score and a total modifiability score are calculated. Specific criteria are delineated for each category. Depending on a child's score they are classified as having *typical learning characteristics; mild, moderate, severe or profound learning challenges* under the learning strategies checklist. They are then rated on the modifiability scale as having observed *typical learning patterns, or atypical mild, moderate or severe learning patterns*. While these scores are used to determine whether an impairment exists, and what the severity of that impairment is, the clinician should not confine themselves to the posttest performance of the child as change can be quantitative and/or qualitative (Gutierrez-Clellen, Brown, Robinson-Zañartu, & Conboy, 1998).

Conclusion

Dynamic assessment is a process-oriented descriptive assessment that uses observable behavior to quantify a child's readiness for learning. The dynamic assessment process provides the clinician with information that informs planning and treatment that go far above and beyond the results of any norm-referenced test. While it undeniably takes more time and energy to implement than traditional standardized assessment measures, it ultimately results in more accurate identification.

One of the largest advantages of dynamic assessment is that its inherent nature provides the clinician with a minimum of three interactions to observe and build rapport with the child: the test, the MLE and the retest. Many times more than one MLE is completed, providing even more opportunities to connect with the child. The client-clinician

relationship begins developing before treatment has begun and time in the therapy room automatically becomes more effective as a result.

Most powerfully, dynamic assessment can prevent children who have a language difference – not a disorder – from entering the special education system. This ultimately results in reduced caseloads, giving clinicians more time to serve the children in need.

Case Examples:

We have supplemented the above description of the dynamic assessment process with five case studies and accompanying videos. It is our goal that these real-life examples will make the process of administering and scoring a dynamic assessment accessible to both graduate students and practicing clinicians. Along with the videos, we have provided written analyses of each case, including detailed background information and step-by-step descriptions of the dynamic assessment, including target selection, administration, scoring and analysis. The videos and written analyses are intended to be used together.

In total, the video examples are approximately an hour long. The videos are meant to provide a visual example of how dynamic assessment “looks” in clinical practice. For each case, we include footage of the pretest, mediated learning experiences, and posttest. The footage is accompanied with voice overs that explain techniques and pose questions to the viewer regarding the assessment. The videos will be most valuable if one is already familiar with the information on this handout and it is particularly useful to have copies of the GDAP scale and Dynamic Assessment Scoring Criteria in front of you as you watch them. Although the case examples provide an in-depth written description of the assessment and our analysis, one can watch the videos before reading the reports in order to evaluate their own reliability. It is our goal that students and clinicians will be more comfortable administering a dynamic assessment for a variety of skills after reviewing the videos.

The case study examples are:

1. **Case Study One** involves a school-aged Spanish-English bilingual who demonstrates how a child with a language disorder presents during a dynamic assessment of narrative ability.
2. **Case Study Two** also involves a school-aged Spanish-English bilingual who demonstrates of a bilingual child with typical learning.
3. **Case Study Three** is a preschool-aged Spanish-English bilingual who demonstrates typical phonological development relative to language-matched peers.
4. **Case Study Four** is a school-aged German-English bilingual who demonstrates atypical phonological development relative to language-matched peers.
5. **Case Study Five** is a Kindergarten-aged German-English bilingual who demonstrates typical language learning in spite of significant teacher concerns.

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Appendix A:

Dynamic Assessment Scoring Sheet, page 1

Learning Strategy Checklist	Scoring Criteria (Circle One)	Comments
Attention Student exhibits an orienting response and focus to task-relevant stimuli	1 2 3 4 5	
Task Specific Performance Student performs features of the target task (e.g., answer questions, describing, sequencing, use comparative relationships, summarizing, etc.)	1 2 3 4 5	
Planning Student verbalizes or uses systematic strategy to task	1 2 3 4 5	
Self Regulation / Awareness Student waits for instructions, seeks help when needed, corrects self, or rewards self	1 2 3 4 5	
Motivation Student shows enthusiasm for task & persists in the face of frustration	1 2 3 4 5	
Interaction with Adults Seeks help when needed & changes responses according to adult cues	1 2 3 4 5	

Total Learning Strategy Score _____ **(Use score for Analysis on Reverse Side)**

Modifiability Scale (Note: overall performance after each teaching session)

Examiner Effort	Extreme (1)	High – Moderate (2)	Moderate-Slight (3)	Slight-None (4)
Student Responsivity	None (1)	Slight (2)	Moderate (3)	High (4)
Transfer	None (1)	Low (2)	Moderate (3)	High (4)

Total Modifiability Score _____ (Use Score for Analysis on Reverse Side)

Differentiation Criteria**Learning Strategies Checklist:**

- 6 Points **Profound** Learning Challenges
 7-12 Points **Severe** Learning Challenges
 13-18 Points **Moderate** Learning Challenges
 19-24 Points **Mild** Learning Challenges
 25-30 Points **Typical** Learning Characteristics (i.e., Learn, Retain, & Transfer Information)

Modifiability Scale:

- 3 Points Observed Learning Patterns - Atypical & **Severe**
 4-6 Points Observed Learning Patterns - Atypical & **Moderate**
 7-9 Points Observed Learning Patterns - Atypical & **Mild**
 10-12 Points Observed Learning Patterns - **Typical**

Note: To maximize the validity of the Dynamic Assessment results, the teaching lessons need to be within the student's "zone of proximal development" (i.e., within their developmental range). If a targeted lesson is too easy or challenging, the results of the Dynamic Assessment can be compromised.

Learning Strategies Checklist

Attention: The student exhibits an orienting response and focus to task-relevant stimuli (i.e., ignoring irrelevant stimuli), throughout length of task. Student who is distractible may need greater repetition of instruction and may require more mediation to learn.

5. Orienting response and on-task behavior without prompts.
4. Attentional prompts needed only at the beginning of mediation session.
3. Prompting / repetition needed less than 50% of the time.
2. Prompting / repetition needed more than 50% of the time
1. High distractibility and minimal attention.

Task Specific Performance: Teaching strategies are used to help a student develop metacognitive and metalinguistic skills related to a task. The examiner may engage the student with concepts pertaining to who, what, when, where, and why related questions. In addition, the examiner should introduce “what if” questions to facilitate thinking about alternative strategies and answers. Students may also discuss with the examiner the relative benefits of alternative scenarios or strategies. This type of teaching encourages independent, hypothetical thinking and learning.

5. Use of targeted features with minimal difficulty and prompting.
4. Use of targeted features between 50-90% of the time with prompting or use of responses that are close to the target but still inaccurate
3. Use of inaccurate responses between 25-50% of the time with prompting.
2. Use of guesses or irrelevant responses.
1. No response.

Planning: The student verbalizes or uses a systematic strategy to engage or complete targeted task.

5. Use or verbalization of systematic strategy to task with minimal difficulty or prompting.
4. Use or verbalization of systematic strategy to task between 50-90% of the time with prompting.
3. Use or verbalization of systematic strategy to task between 25-50% of the time with prompting.
2. Use or verbalization of systematic strategy to task less than 25% of the time with prompting.
1. No systematic strategy used for task.

Self-Regulation: The student demonstrates self-awareness and self-regulation skills. For example, the student waits for instructions, asks for assistance when task becomes difficult, responds to instruction with minimal prompts, is aware of correctness of own responses, and may use self-rewarding behavior.

5. Use of self-corrections and self-rewards for accurate response consistently after initial feedback.
Waits for instruction throughout session.
4. Use of self-correction and self-reward between 50-90% of the time. Waits for instruction throughout the session.
3. Minimal use of self-correction or self-reward. Occasional impulsive response.
2. Impulsivity and fidgety 50% of the time.
1. Impulsivity and fidgety throughout the session.

Motivation: The student shows enthusiasm for the task and persists in the face of frustration.

5. High enthusiasm about task as noted by verbalization, body language, and/or facial expressions.
4. Willingness to try most tasks throughout the session, despite difficulty.
3. Occasional attempts to end mediation activities.
2. Frequent attempts to end mediation activities.
1. Refusal to participate in mediation activities.

Interaction with Adults: The student seeks help when the task becomes difficult. The student changes responses according to adult cues and feedback.

5. Consistent use of verbal / nonverbal requests for help when task becomes difficult. Response consistently changes based on adult input.
4. Use of requests for help most of the time. Occasional repetition of old errors in spite of adult input.
3. Inconsistent change in responses despite adult feedback (<50% of the time).
2. Frequent repetition of old errors despite adult feedback (> 50% of the time).
1. Minimal change in response to adult feedback.

Modifiability Scale

Examiner Effort: Using the data from the learning strategies checklist, identify the amount and intensity of effort required to induce change during the teaching session.

4. Slight - None
3. Moderate – Slight
2. High – Moderate
1. Extreme

Student's Response to Teaching: Using the data from the learning strategies checklist, rate the student's overall responsiveness to the examiner's teaching during the lesson.

4. High
3. Moderate
2. Slight
1. None

Ability to Transfer: Using the data from the learning strategies checklist, rate how the student applies the learned strategies to a new task.

4. High
3. Moderate
2. Low
1. None

Adapted by Frank Bender from the following sources:

Gutiérrez-Clellen, V., Brown, S., Conboy, B., and Robinson-Zanartu, C. (1998). Modifiability: A dynamic approach to assessing immediate language change. *Journal of Children's Communication Development*, 19 (2), 31-43.

Peña, E. (2000). Measurement of modifiability in children from culturally and linguistically diverse backgrounds. *Communication Disorders Quarterly*, 21 (2), 87-97